POORNIMA COLLEGE OF ENGINEERING

DEPARTMENT OF INFORMATION TECHNOLOGY

MINOR PROJECT PRESENTATION

ON

POLLU-SENSE

AI-Based Air Quality Prediction and Monitoring System

Session: 2026-27

Presented to:
Mr. Arpit Jain
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(Faculty Coordinators)

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Aditi Jain (PCE23IT004)

POLLU-SENSE

- **Problem Title:** AI-powered system for real-time air pollution tracking, notification, and control
- **Problem Description:** This project aims to develop an AI integrated real time Air Pollution Tracking System which keeps a track of potential Air pollution causing agents. Our AI-powered system continuously tracks citywide pollution, identifies spikes, and triggers instant alerts. By integrating IoT and predictive analytics, it ensures accountability, enabling proactive measures for cleaner air.
- The whole Sole motto of this project is to ensure that our future generations get a fresh and clean air to breathe in.

POLLU-SENSE

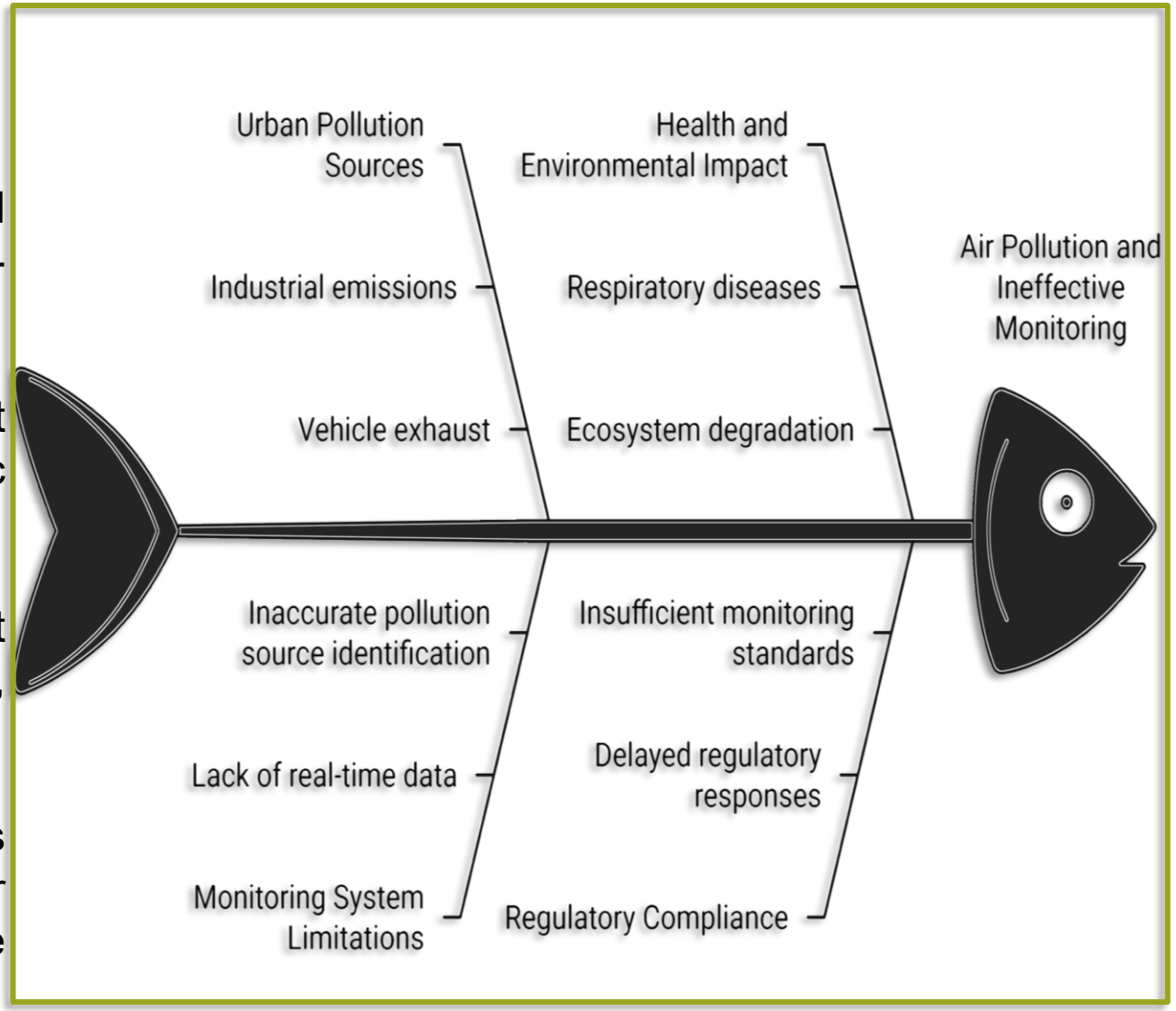
Key Challenges & Pain Points:

•**Lack of Real-Time Monitoring:** Most pollution control boards rely on periodic air quality reports instead of real-time tracking.

•**Inability to Identify Polluters:** Current systems detect high pollution levels but do not attribute them to specific sources.

•**Delayed Action & Compliance Issues:** Without instant alerts, industries and authorities fail to act on time, leading to long-term environmental damage.

•**Public Health Concerns:** Air pollution causes respiratory diseases, cardiovascular issues, and other serious health conditions, affecting vulnerable populations.



POLLU-SENSE

❖ Health Impacts:

Respiratory Infections: Long-term exposure to particulate matter increases the risk of respiratory diseases such as asthma, bronchitis, chronic obstructive pulmonary disease (COPD), and lung infections

•**Cancer:** It contributes to the development of lung cancer and has been linked to other cancers due to the presence of carcinogenic pollutants.

•**Other Diseases:** Exposure to air pollution is associated with cardiovascular diseases, adverse pregnancy outcomes, and exacerbation of pre-existing health conditions.

❖ Regional Impact:

Delhi, Mumbai, and Indo-Gangetic Plain face severe pollution due to factors like industrial activities, vehicular emissions, and agricultural practices such as stubble burning.

❖ Economic Impact:

Air pollution leads to substantial economic losses in India, including increased healthcare costs, reduced productivity, and loss of labor income due to premature deaths.

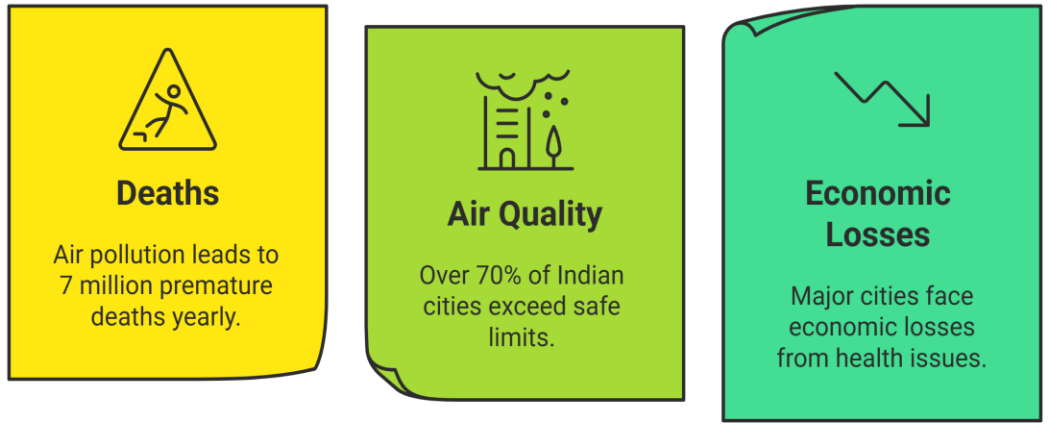
❖ Supporting Data:

•WHO reports that air pollution causes 7 million premature deaths annually.

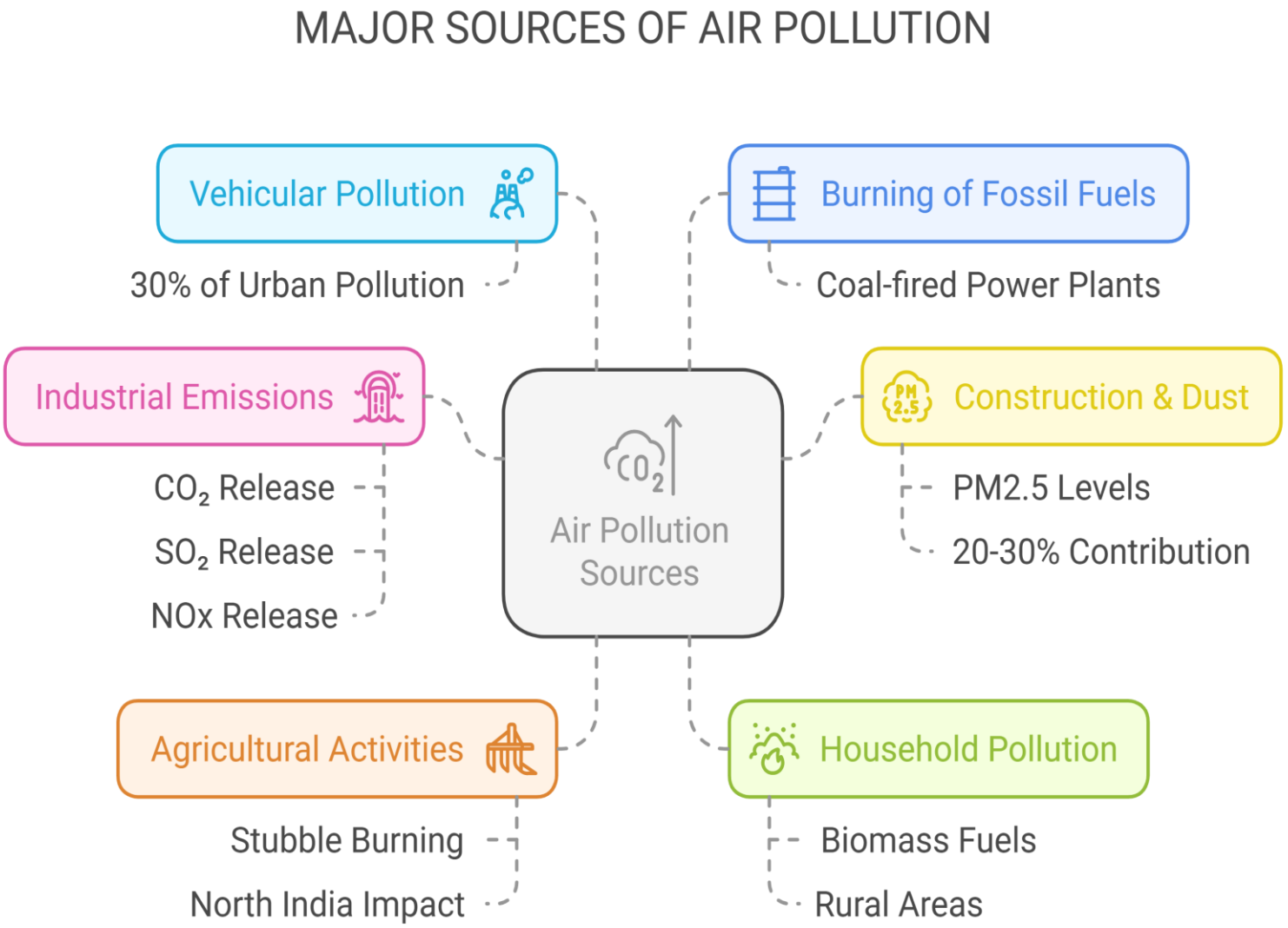
•A 2023 CPCB report states that over 70% of Indian cities exceed safe air quality limits.

•India loses $150 billion (3% of GDP) annually due to pollution-related health issues.

•Cities like Delhi, Mumbai, and Kolkata experience annual economic losses due to pollution-related health issues and reduced workforce productivity.



POLLU-SENSE



❖ Major Causes of Air Pollution

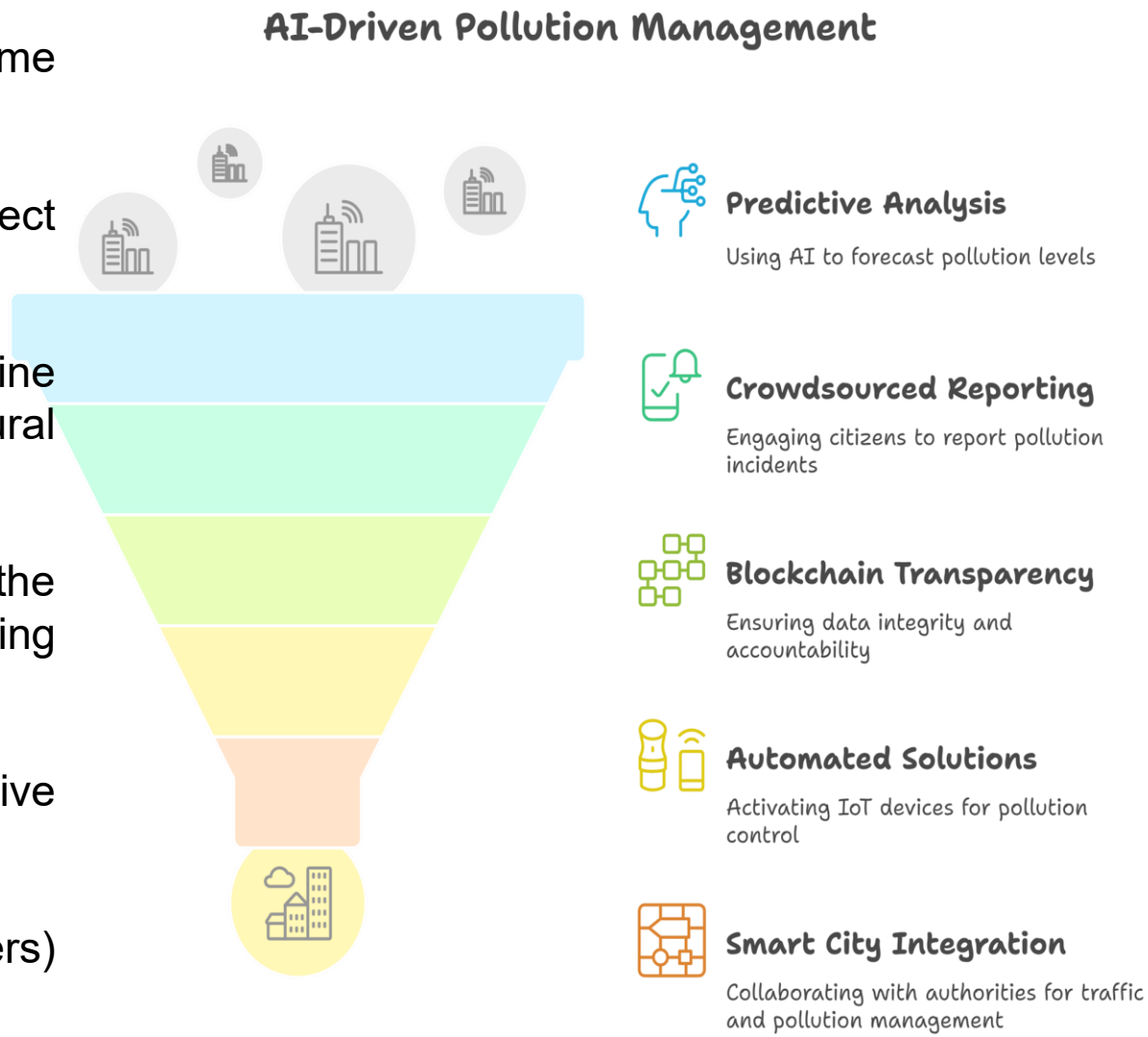
- **Industrial Emissions** – Factories release CO_2, SO_2, and NOx into the air.
- **Vehicular Pollution** – Transportation contributes 30% of total air pollution in cities.
- **Construction & Dust** – 20-30% of PM2.5 levels in urban areas come from construction.
- **Burning of Fossil Fuels** – Coal-fired power plants are a major contributor.
- **Agricultural Activities** – Stubble burning increases air pollution, especially in North India.
- **Household Pollution** – Use of biomass fuels for cooking in rural areas.

POLLU-SENSE

❖ Proposed Solution:

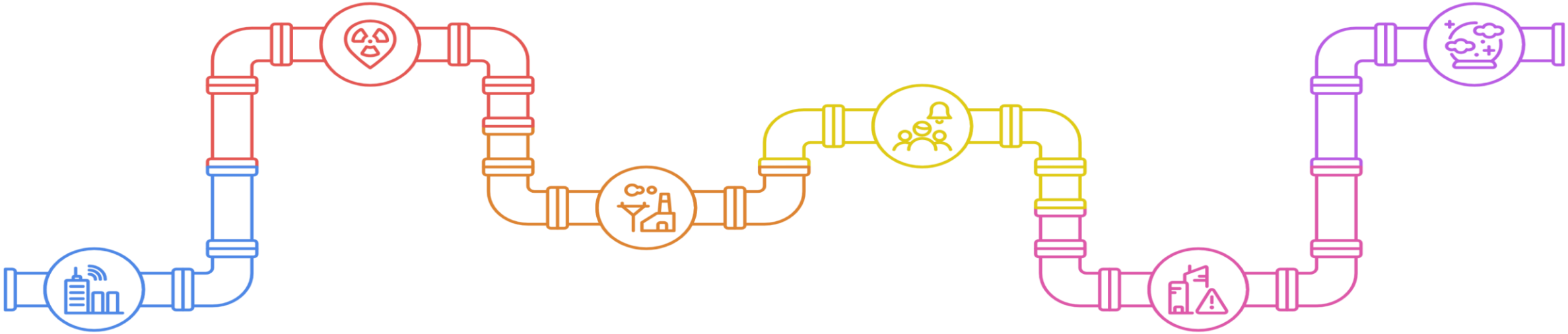
Our **AI-based Pollution Tracking System** integrates real-time monitoring, smart notifications, and automated action plans:

- **Real-Time AI Detection:** Uses sensors and satellite data to detect pollution spikes in different regions.
- **Source Identification:** AI algorithms analyze data to determine whether pollution originates from traffic, industries, or natural causes.
- **Automated Notifications:** If pollution exceeds a threshold, the system alerts responsible industries or individuals, prompting immediate corrective action.
- **Escalation Mechanism:** If no action is taken, authorities receive reports for regulatory intervention.
- **IoT-Powered Solutions:** Smart air purifiers (mist sprays, filters) activate in high-pollution areas for immediate impact.



FLOW-CHART

AI-Powered Pollution Tracking System



01

Monitor Air Quality

Continuous monitoring of air quality across city regions

02

Detect Pollution Spike

AI detects when pollution levels exceed a threshold

03

Identify Pollution Source

AI identifies the source of the pollution spike

04

Notify Responsible Party

Responsible individuals or industries are notified

05

Escalate to Authorities

Issue escalated to government authorities if necessary

06

Predict Future Levels

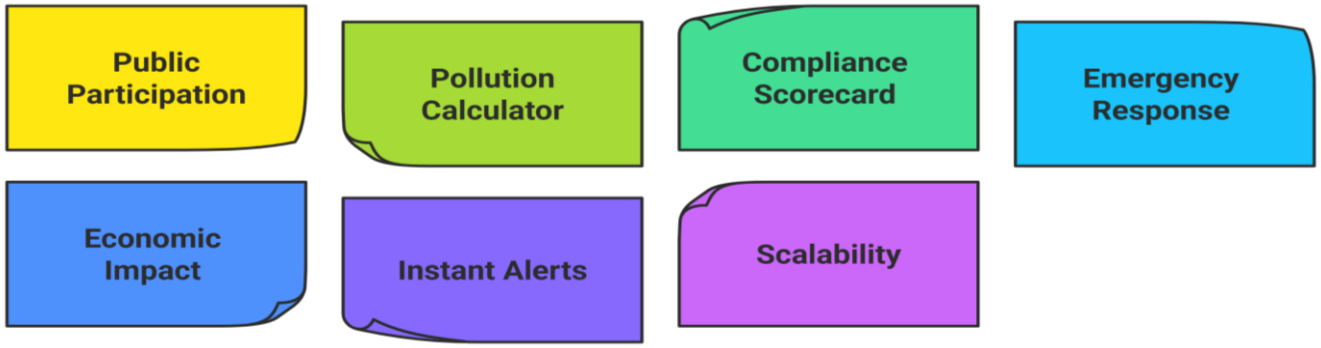
AI predicts future pollution levels using data

❖ IMPACT

- ✓ AI predicts future pollution spikes, enabling preventive measures before air quality worsens.
- ✓ Identifies specific polluters (factories, vehicles, or construction sites) and alerts them to take corrective action.
- ✓ Crowdsourced reporting allows citizens to contribute real-time pollution data, increasing transparency.
- ✓ Aligns with smart city initiatives by integrating with traffic management, environmental monitoring, and urban planning systems.

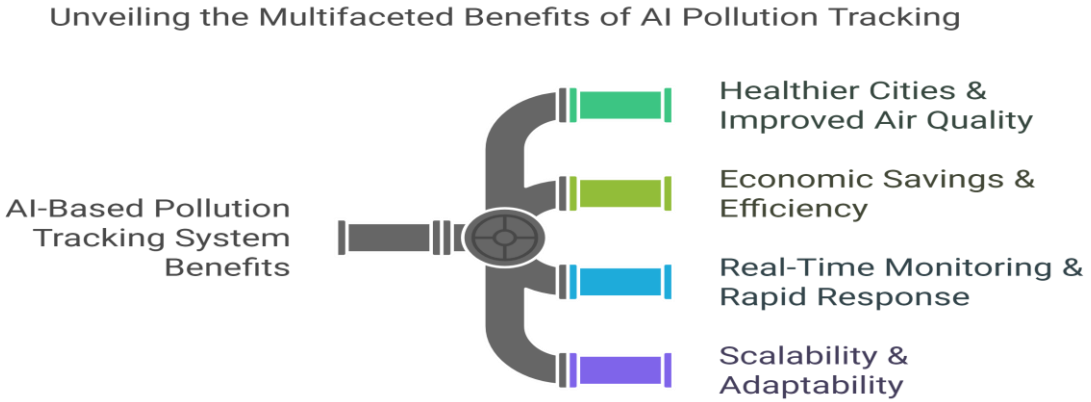
❖ BENEFITS

- ✓ Automated air purification reduces exposure to harmful pollutants, lowering respiratory diseases and healthcare costs.
- ✓ Reduces economic losses from pollution-related health issues and workforce productivity drops.
- ✓ Instant alerts allow industries and authorities to act quickly, minimizing long-term pollution impact.
- ✓ If an industry resumes polluting after clearing an inspection, Our AI will alert authorities immediately, ensuring action is taken.
- ✓ Can be expanded to multiple cities and adapted to different pollution sources (industrial, vehicular, or natural).

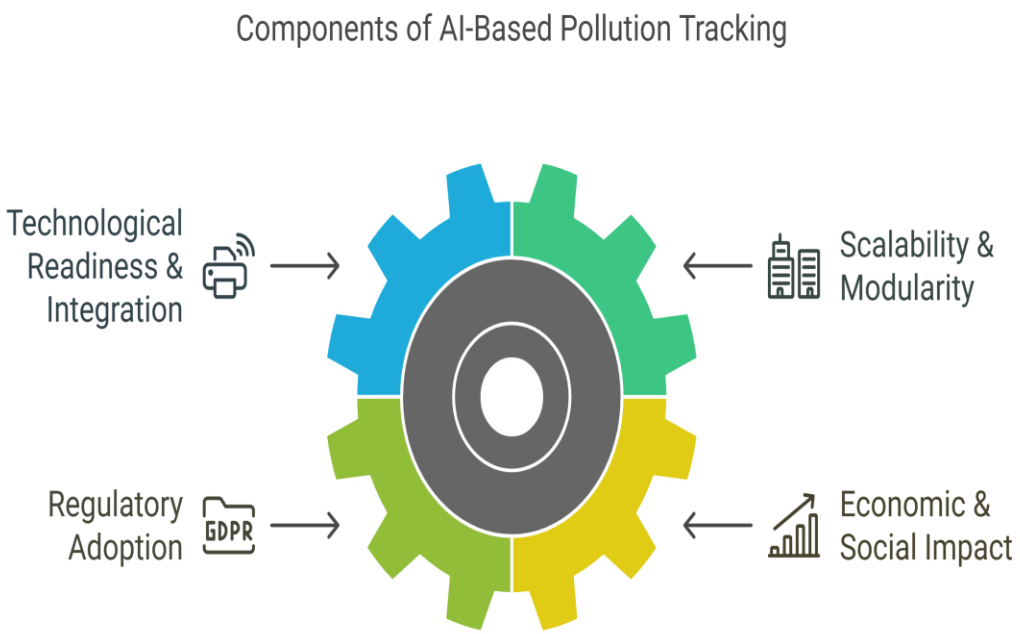


❖ FEATURES

- ✓ **Credified Public Participation System:** Citizens earn rewards for reporting pollution sources or using public transport.
- ✓ **AI-Based Pollution Footprint Calculator:** Tracks personal exposure and suggests eco-friendly choices based on daily habits.
- ✓ **Automated Industry Compliance Scorecard:** Assigns environmental ratings to industries based on real-time and historical data.
- ✓ **Emergency Response Triggers for High Pollution Alerts:** AI activates roadside mist sprays and air filters in critical zones.



POLLU-SENSE



❖ FEASIBILITY & VIABILITY

- **Technological Readiness & Integration:** AI-powered pollution detection is achievable using existing air quality sensors, IoT devices, and cloud-based analytics, making implementation feasible.
- **Scalable & Modular System:** The system can start in highly polluted cities and later expand, integrating new sensors, data sources, and smart city infrastructure as needed.
- **Regulatory & Smart City Adoption Potential:** Governments and municipalities can adopt it as part of pollution control policies and smart city initiatives, ensuring long-term sustainability.
- **Economic & Social Impact:** Reduces healthcare costs, improves workforce productivity, and enhances quality of life, making it economically viable for industries and urban planners.

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❖ROADMAP & TECHSTACK

1.Data Collection & Sensor Integration: Use OpenWeatherMap API, CPCB data, and IoT sensors for real-time pollution tracking.

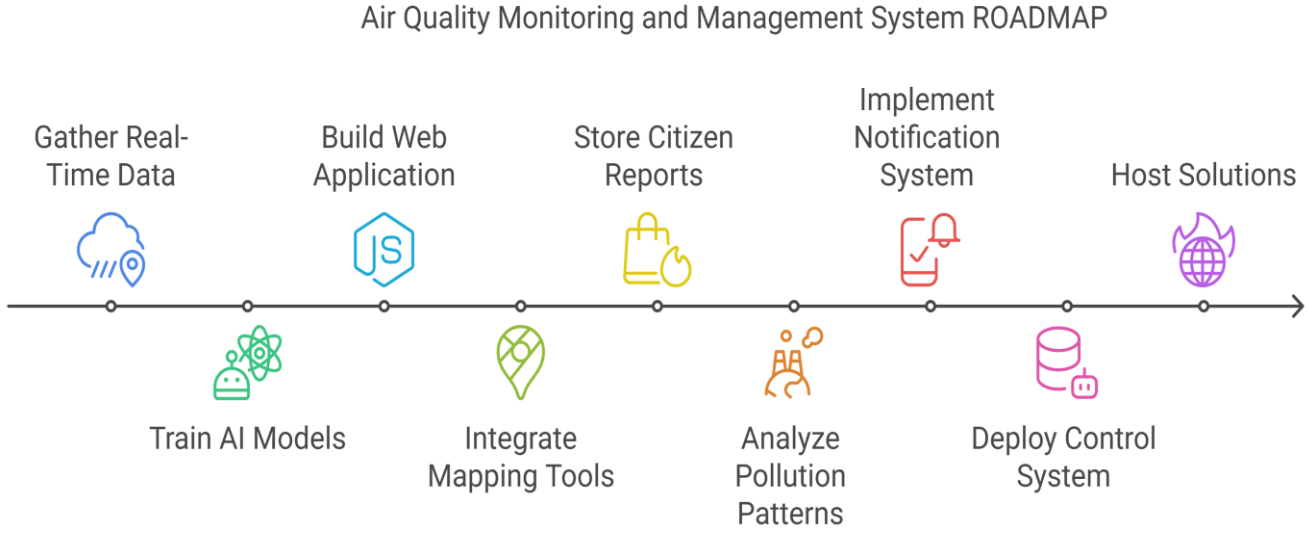
2.Backend Development & AI Model Training: Implement Django (Python), TensorFlow, and Scikit-learn for AI-driven pollution predictions.

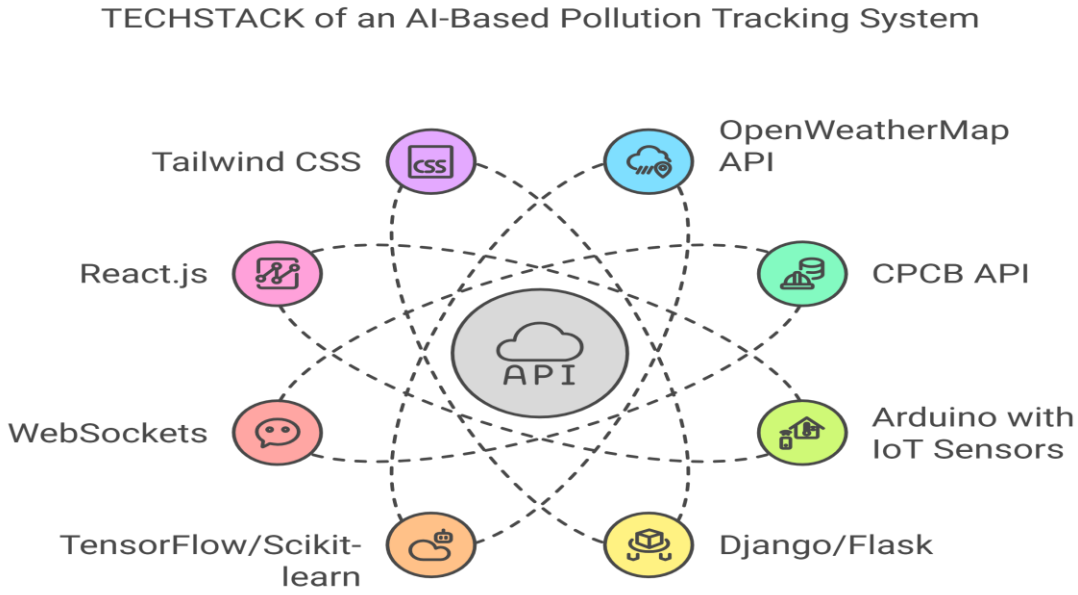
3.Frontend & User Interaction Platform: Build a React.js website with Firebase authentication and crowdsourced reporting features.

4.Real-Time Monitoring & Alerts System: Deploy Flask API with WebSockets for instant notifications using Firebase Cloud Messaging.

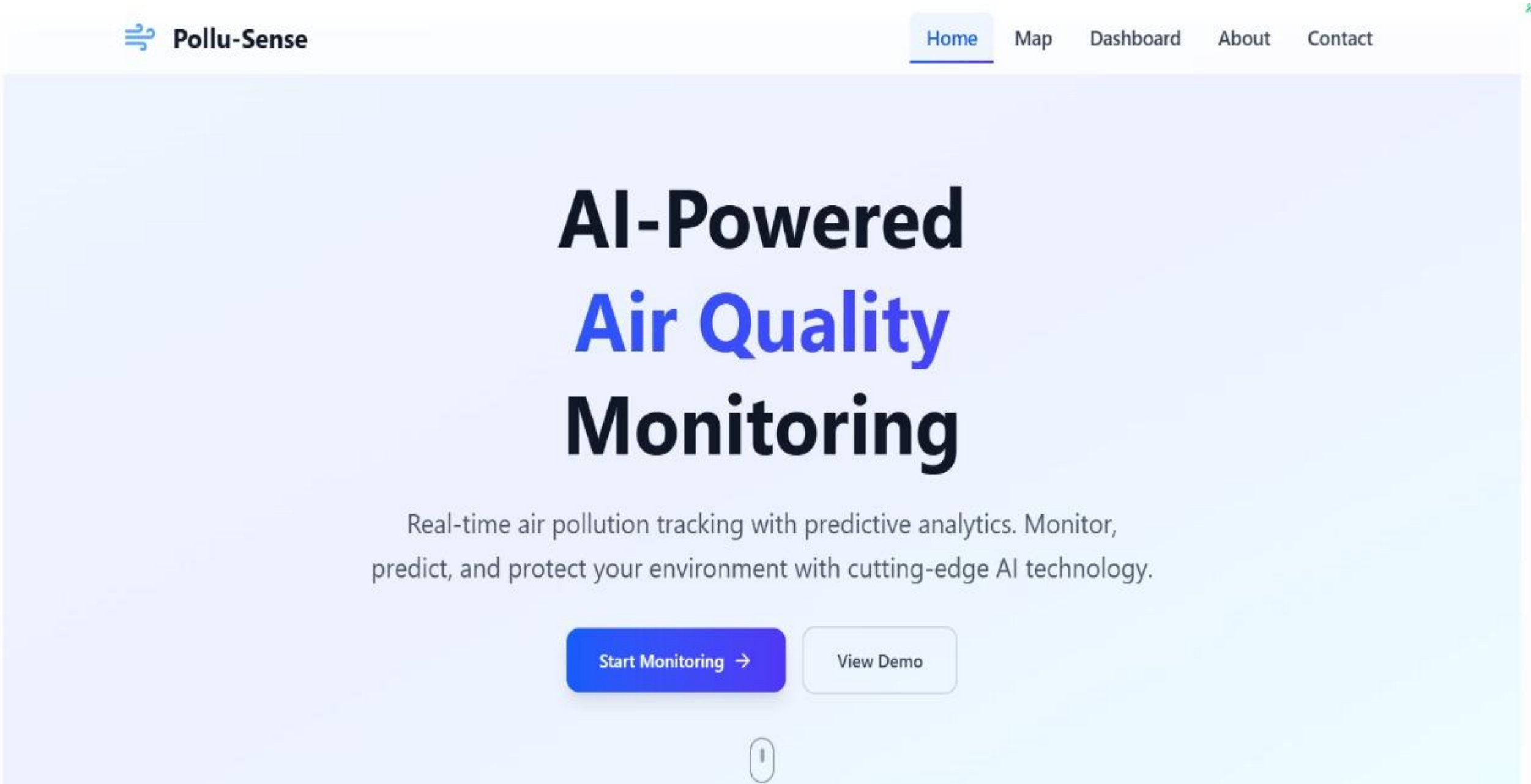
5.Industry Compliance & Historical Tracking Module: Store pollution patterns in PostgreSQL database, visualized through Chart.js dashboard.

6.Final Deployment & Smart City Integration: Host on Render (free tier), integrate with Google Maps API for visualization.





SCREENSHOTS:-



How It Works

Our advanced AI system processes real-time data to deliver accurate predictions



IoT Sensors

Deploy smart sensors across the city to collect real-time air quality data



AI Processing

Advanced machine learning algorithms analyze patterns and trends



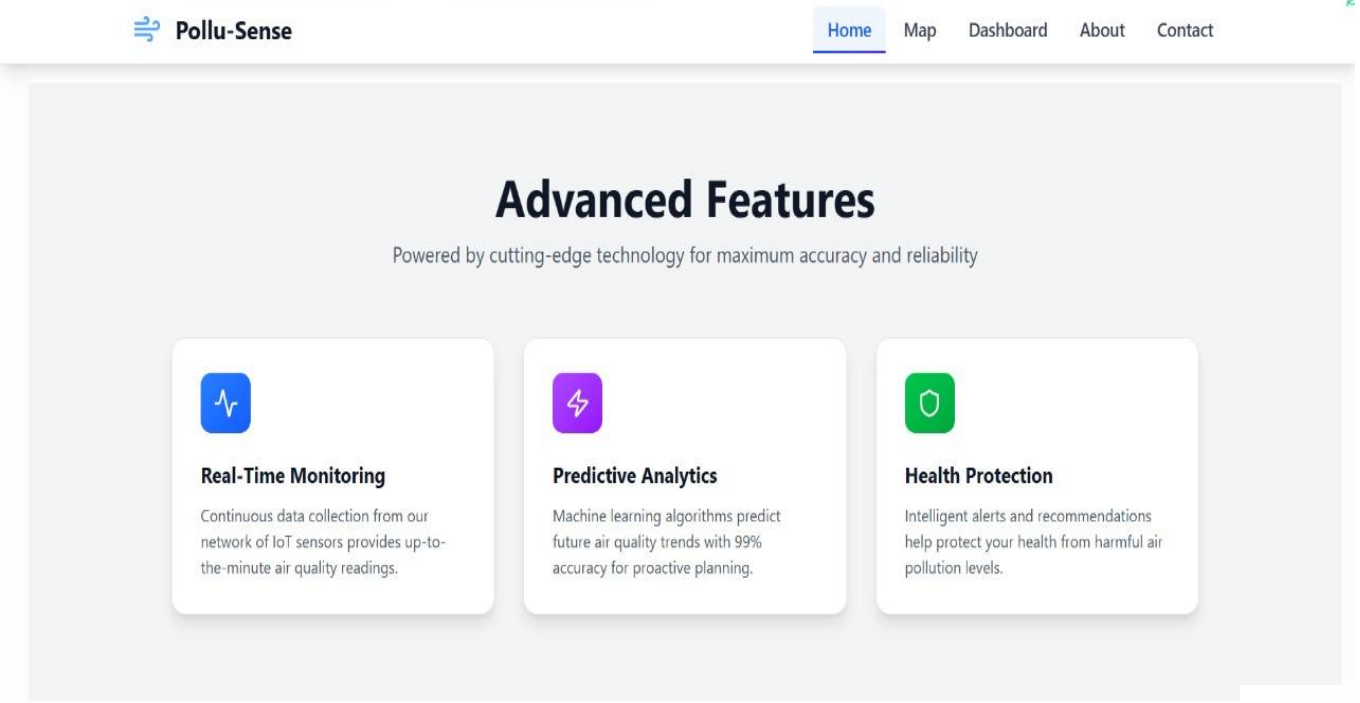
Predictions

Generate accurate forecasts and pollution trend predictions



Smart Alerts

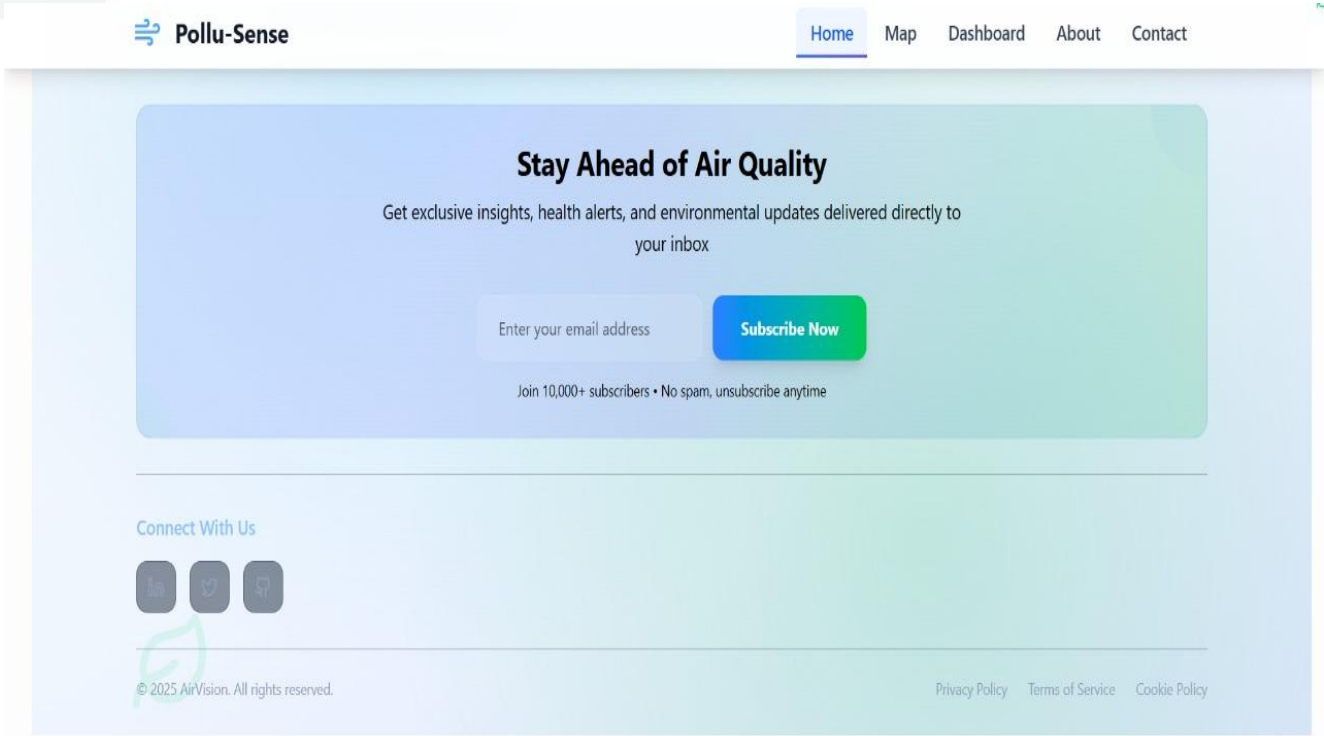
Receive instant notifications when air quality changes

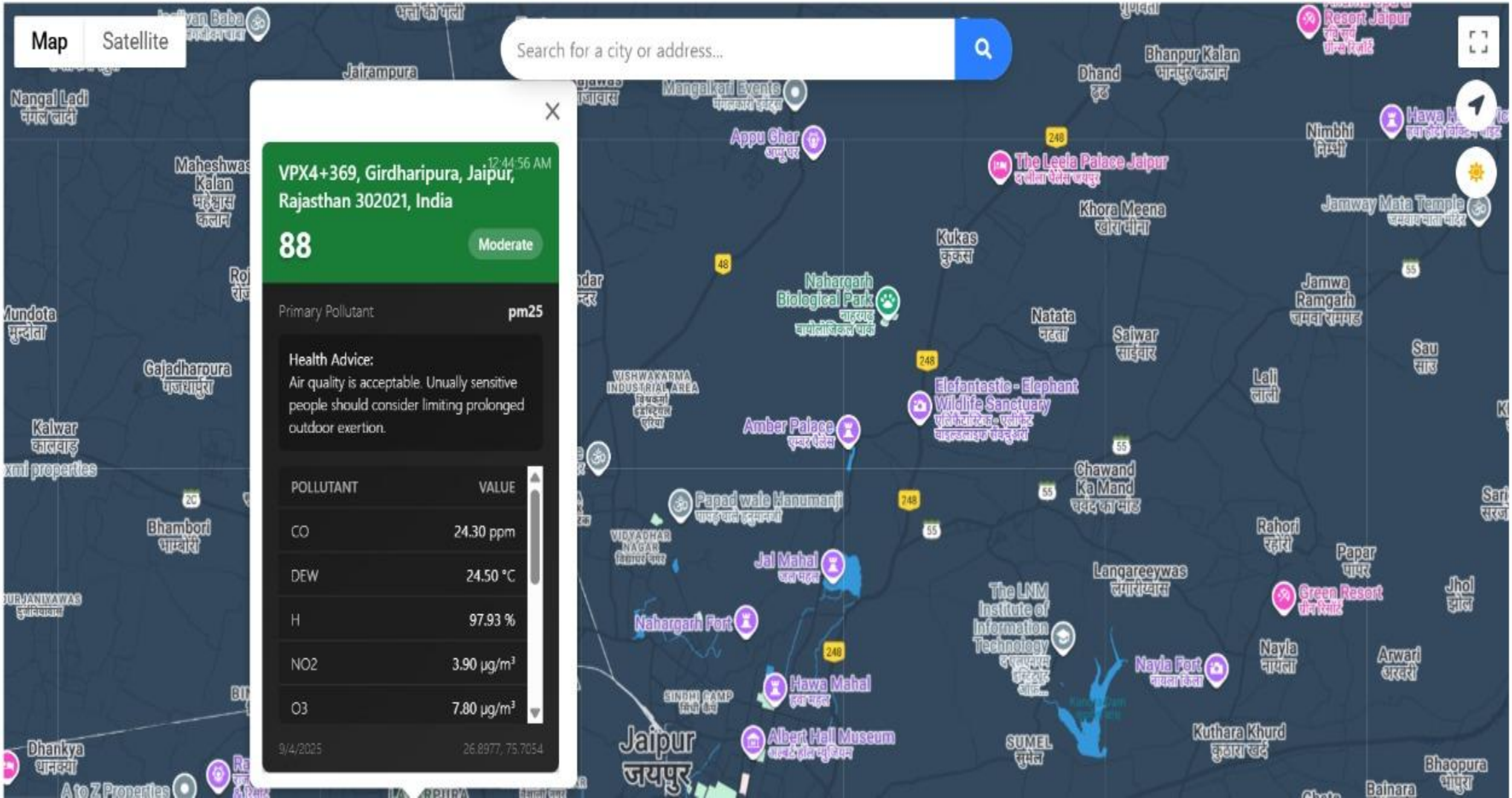


Features:-

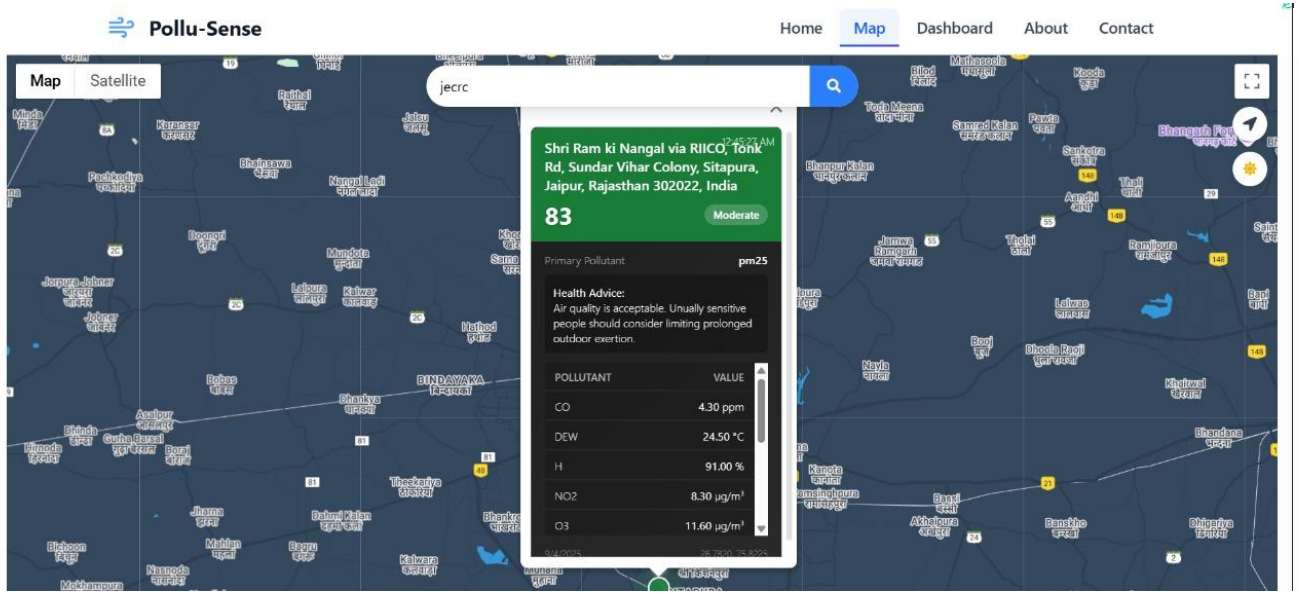
- Real-time Monitoring
- Predictive Analysis
- Health Protection

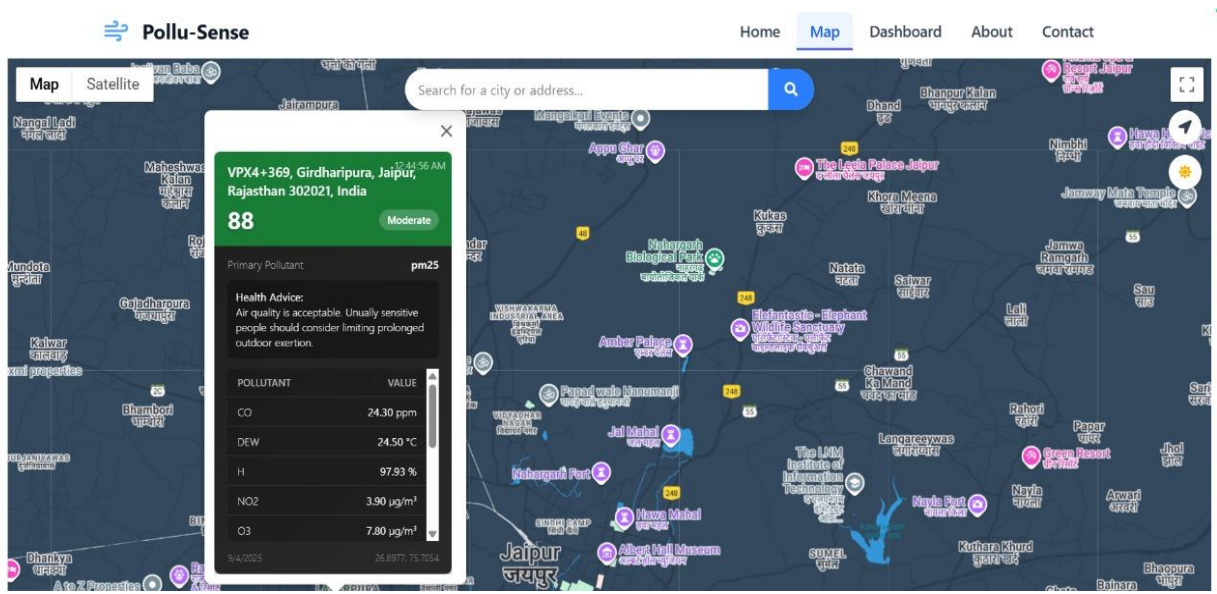
News Letter Subscription

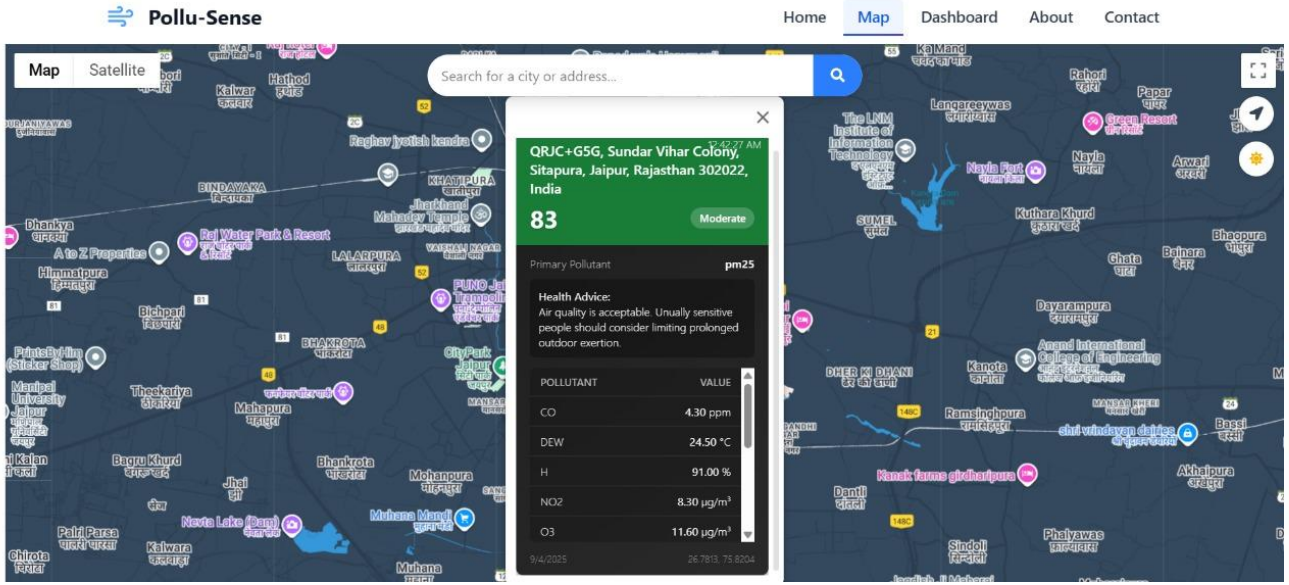


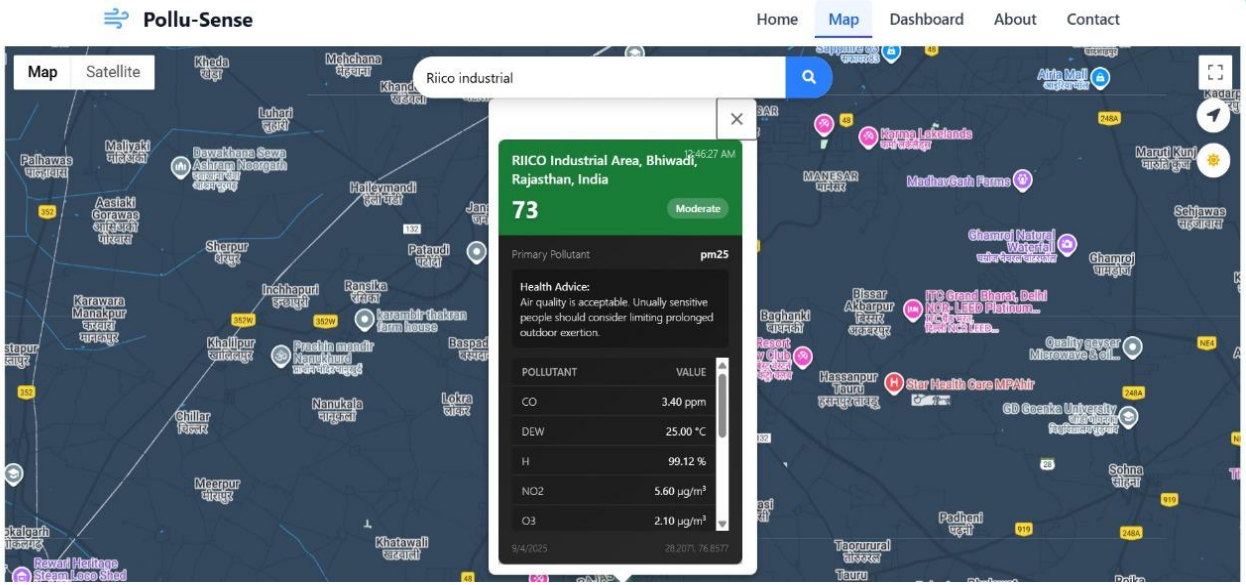

Map
Satellite
Search for a city or address...
VPX4+369, Girdharipura, Jaipur, Rajasthan 302021, India
12:44:56 AM
88
Moderate
Primary Pollutant
pm25
Health Advice:
Air quality is acceptable. Unually sensitive people should consider limiting prolonged outdoor exertion.
POLLUTANT
VALUE
CO
24.30 ppm
DEW
24.50 °C
H
97.93 %
NO2
3.90 µg/m³
O3
7.80 µg/m³
9/4/2025
26.8977, 75.7054
Jaipur
जयपुर
Nangal Ladi
Jairampura
Maheshwas Kalan
Mundota
Gajadharpura
Kalwar
Bhambori
Dhankya
Mangalkari Events
Appu Ghar
Dhand
Bhanpur Kalan
Resort Jaipur
Nimbhi
The Leela Palace Jaipur
Khora Meena
Jamway Mata Temple
Kukas
Nahargarh Biological Park
Natata
Saiwar
Jamwa Ramgarh
Sau
Lali
Elefantastic - Elephant Wildlife Sanctuary
Amber Palace
Chawand Ka Mand
Papad wale Hanumanji
Jal Mahal
Rahori
Papar
Jhol
Langareeywas
Green Resort
Nahargarh Fort
The LNM Institute of Information Technology
Nayla
Nayla Fort
Arwari
Hawa Mahal
Albert Hall Museum
SUMEL
Kuthara Khurd
Bhaopura
Bainara
A to Z Properties
VISHWAKARMA INDUSTRIAL AREA
VIDYADHAR NAGAR
SINDHI CAMP

MAPS:-


Pollu-Sense
Home
Map
Dashboard
About
Contact
Map
Satellite
jecrc
Shri Ram ki Nangal via RIICO, Tonk Rd, Sundar Vihar Colony, Sitapura, Jaipur, Rajasthan 302022, India
83
Moderate
Primary Pollutant
pm25
Health Advice:
Air quality is acceptable. Unually sensitive people should consider limiting prolonged outdoor exertion.
POLLUTANT
VALUE
CO
4.30 ppm
DEW
24.50 °C
H
91.00 %
NO2
8.30 µg/m³
O3
11.60 µg/m³


Pollu-Sense
Home
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Map
Satellite
Search for a city or address...
VPX4+369, Girdharipura, Jaipur, Rajasthan 302021, India
88
Moderate
Primary Pollutant
pm25
Health Advice:
Air quality is acceptable. Unually sensitive people should consider limiting prolonged outdoor exertion.
POLLUTANT
VALUE
CO
24.30 ppm
DEW
24.50 °C
H
97.93 %
NO2
3.90 µg/m³
O3
7.80 µg/m³
Jaipur


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Map
Satellite
Search for a city or address...
QRJC+G5G, Sundar Vihar Colony, Sitapura, Jaipur, Rajasthan 302022, India
83
Moderate
Primary Pollutant
pm25
Health Advice:
Air quality is acceptable. Unually sensitive people should consider limiting prolonged outdoor exertion.
POLLUTANT
VALUE
CO
4.30 ppm
DEW
24.50 °C
H
91.00 %
NO2
8.30 µg/m³
O3
11.60 µg/m³


Pollu-Sense
Home
Map
Dashboard
About
Contact
Map
Satellite
Riico industrial
RIICO Industrial Area, Bhiwadi, Rajasthan, India
73
Moderate
Primary Pollutant
pm25
Health Advice:
Air quality is acceptable. Unually sensitive people should consider limiting prolonged outdoor exertion.
POLLUTANT
VALUE
CO
3.40 ppm
DEW
25.00 °C
H
99.12 %
NO2
5.60 µg/m³
O3
2.10 µg/m³

Air Quality Dashboard

Real-time monitoring of environmental conditions with live sensor data updates and predictive analytics

Last Updated: 9/4/2025, 6:05:30 PM

HUMIDITY
90%

TEMPERATURE
28.9°C

AIR QUALITY INDEX
33
MODERATE

Pollu-Sense

Home Map Dashboard About Contact

Air Quality Dashboard

Real-time monitoring of environmental conditions with live sensor data updates and predictive analytics

Last Updated: 9/4/2025, 6:10:00 PM

HUMIDITY
90%

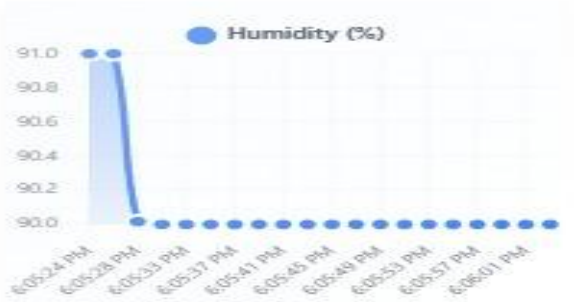
TEMPERATURE
28.9°C

AIR QUALITY INDEX
159
VERY UNHEALTHY

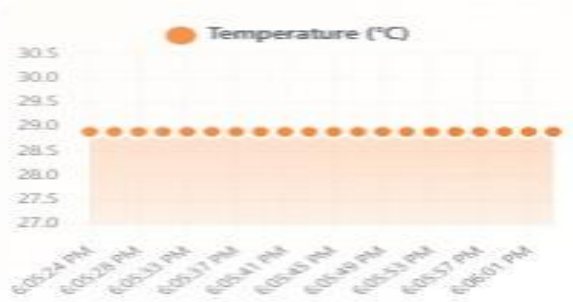
Real-Time Trends

Live sensor data trends for comprehensive environmental monitoring

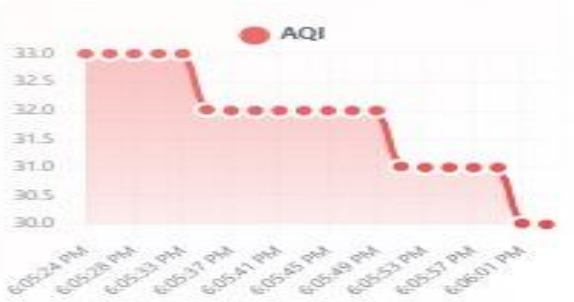
Humidity Trends



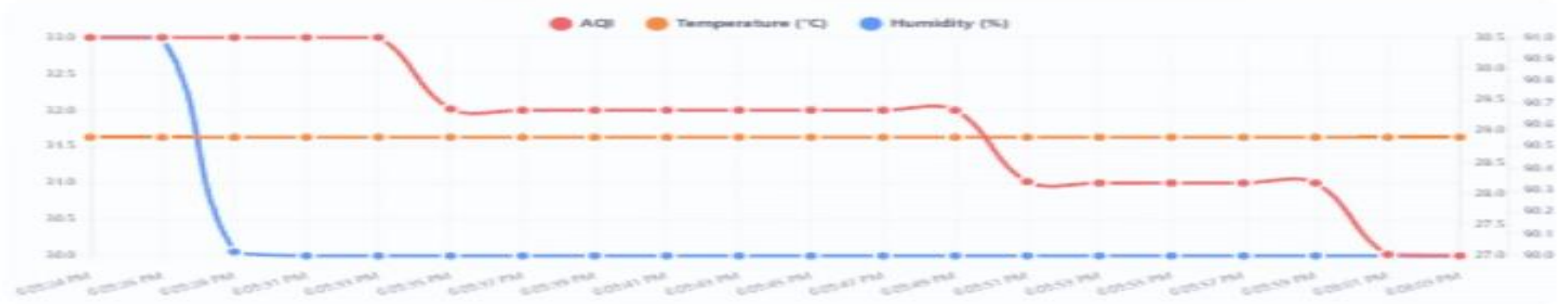
Temperature Trends



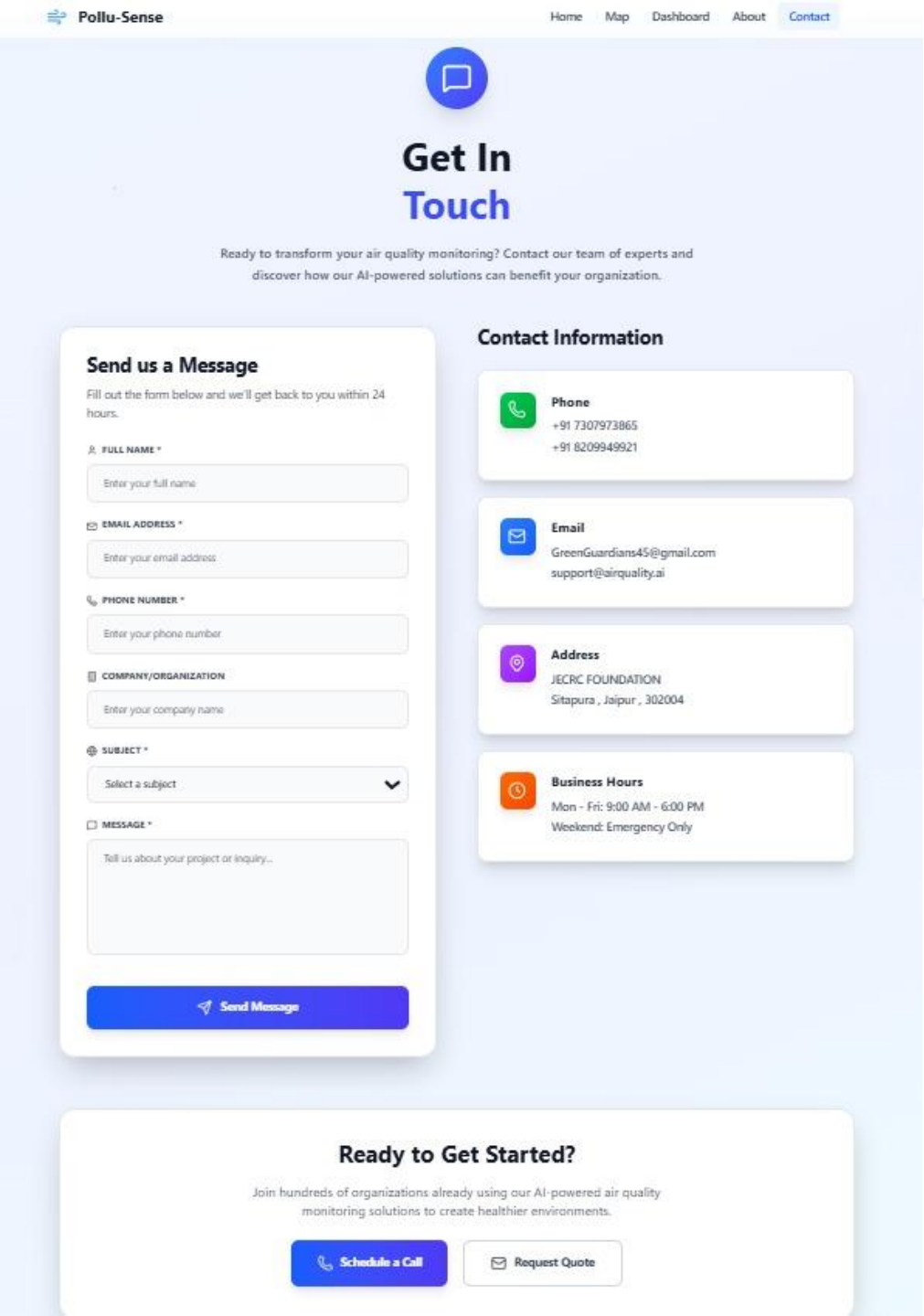
AQI Trends

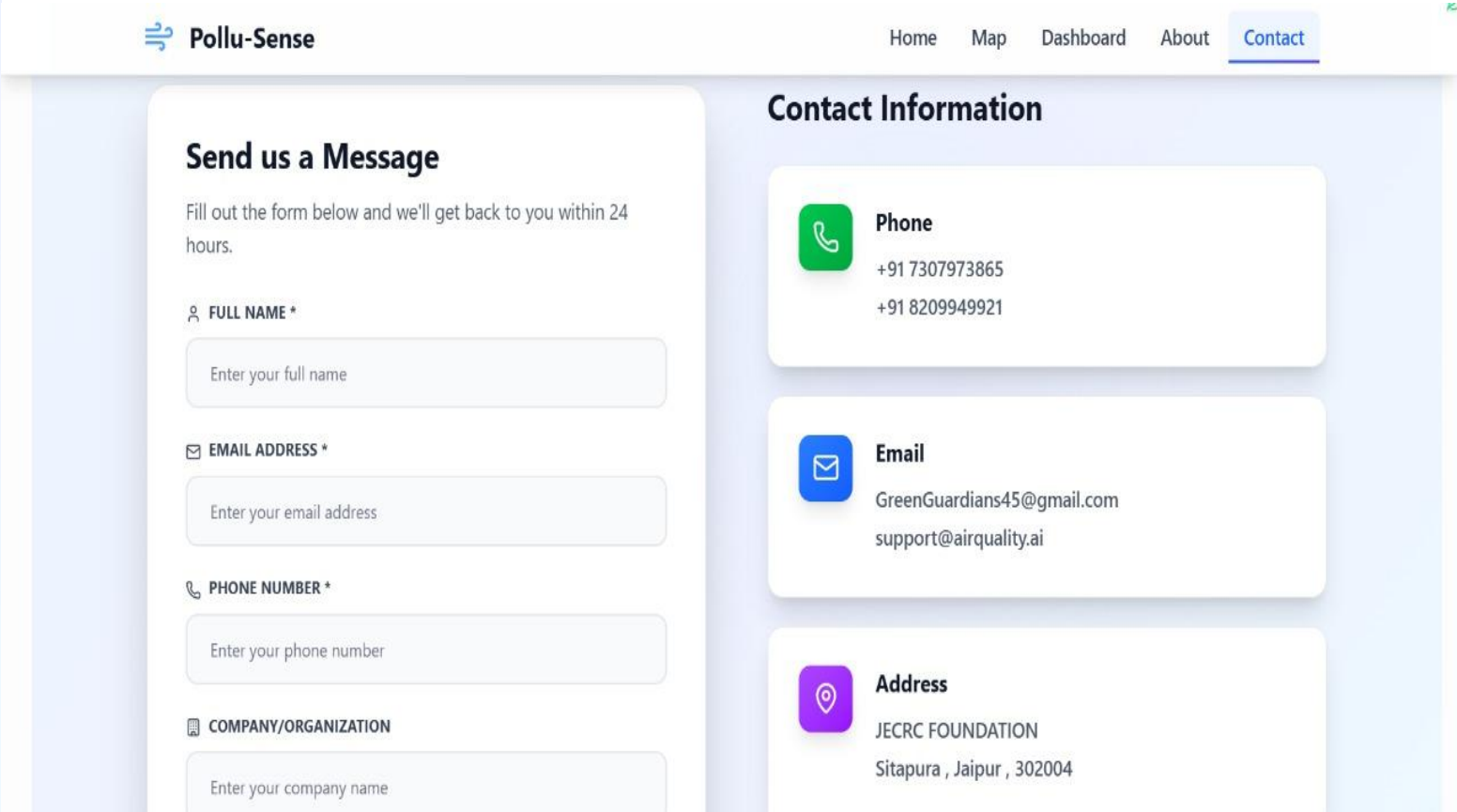


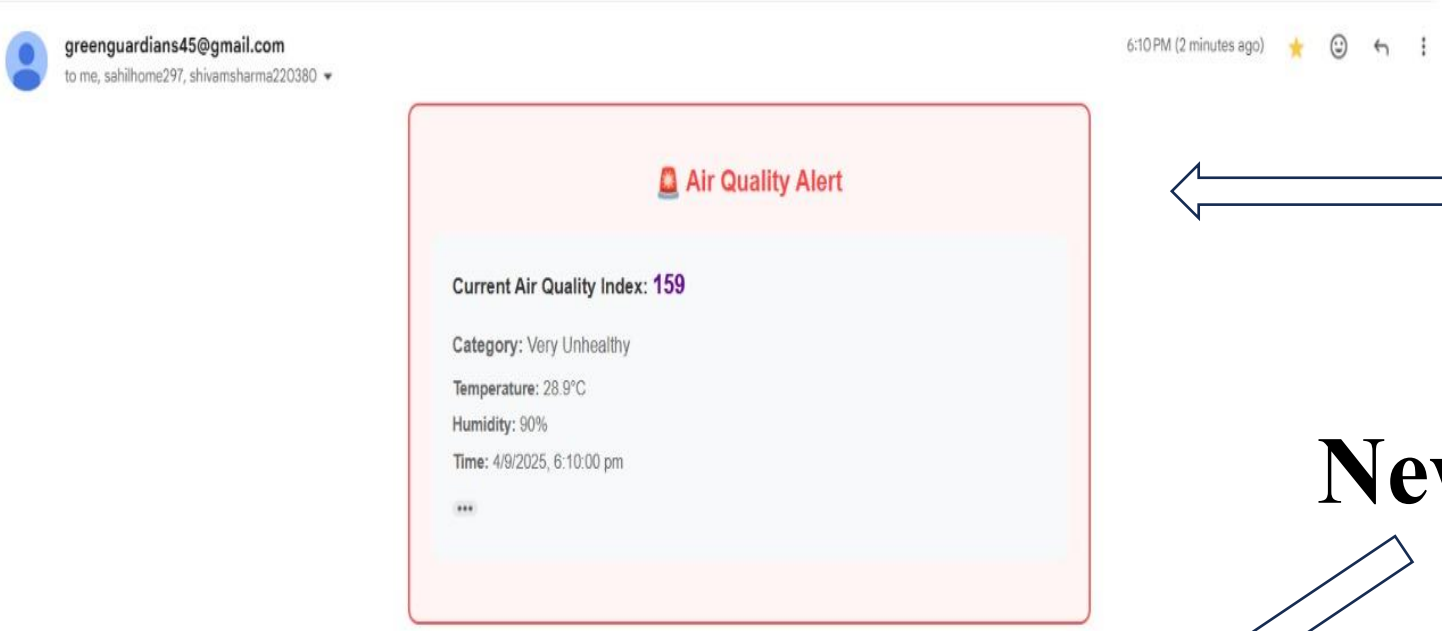
Combined Sensor Data



CONTACT US:-

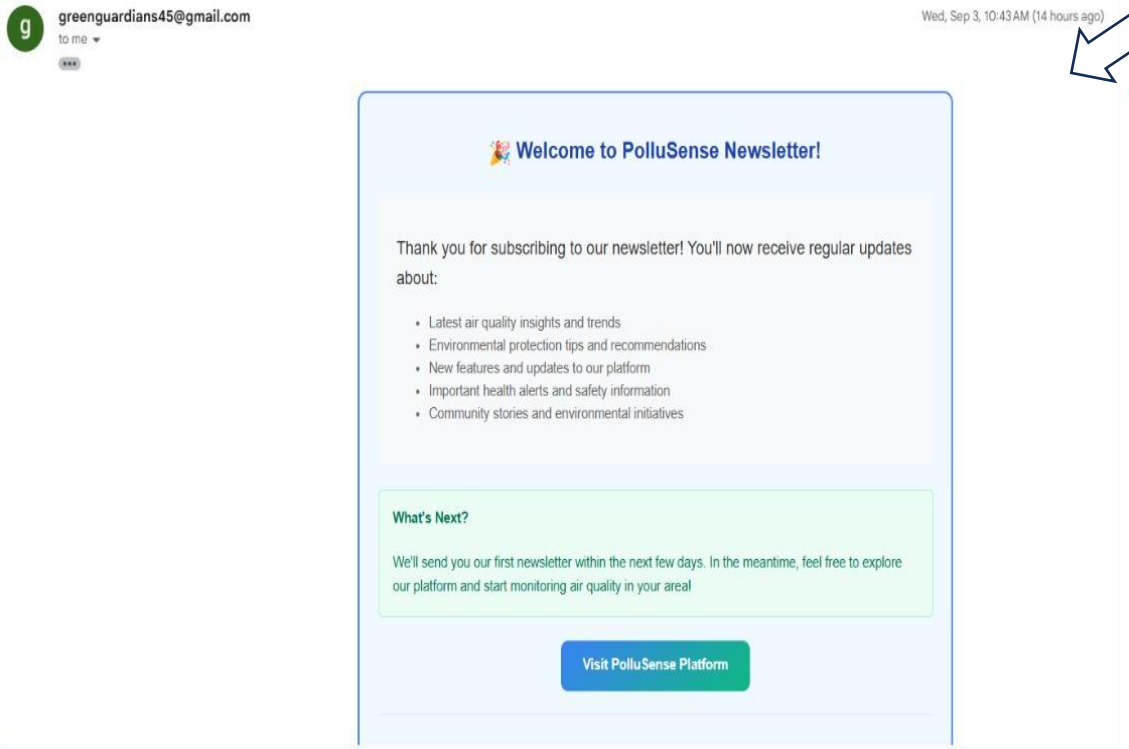

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Ready to transform your air quality monitoring? Contact our team of experts and discover how our AI-powered solutions can benefit your organization.
Send us a Message
Fill out the form below and we'll get back to you within 24 hours.
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Enter your full name
EMAIL ADDRESS *
Enter your email address
PHONE NUMBER *
Enter your phone number
COMPANY/ORGANIZATION
Enter your company name
SUBJECT *
Select a subject
MESSAGE *
Tell us about your project or inquiry...
Send Message
Contact Information
Phone
+91 7307973865
+91 8209949921
Email
GreenGuardians45@gmail.com
support@airquality.ai
Address
JECRC FOUNDATION
Sitapura , Jaipur , 302004
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Mon - Fri: 9:00 AM - 6:00 PM
Weekend: Emergency Only
Ready to Get Started?
Join hundreds of organizations already using our AI-powered air quality monitoring solutions to create healthier environments.
Schedule a Call
Request Quote

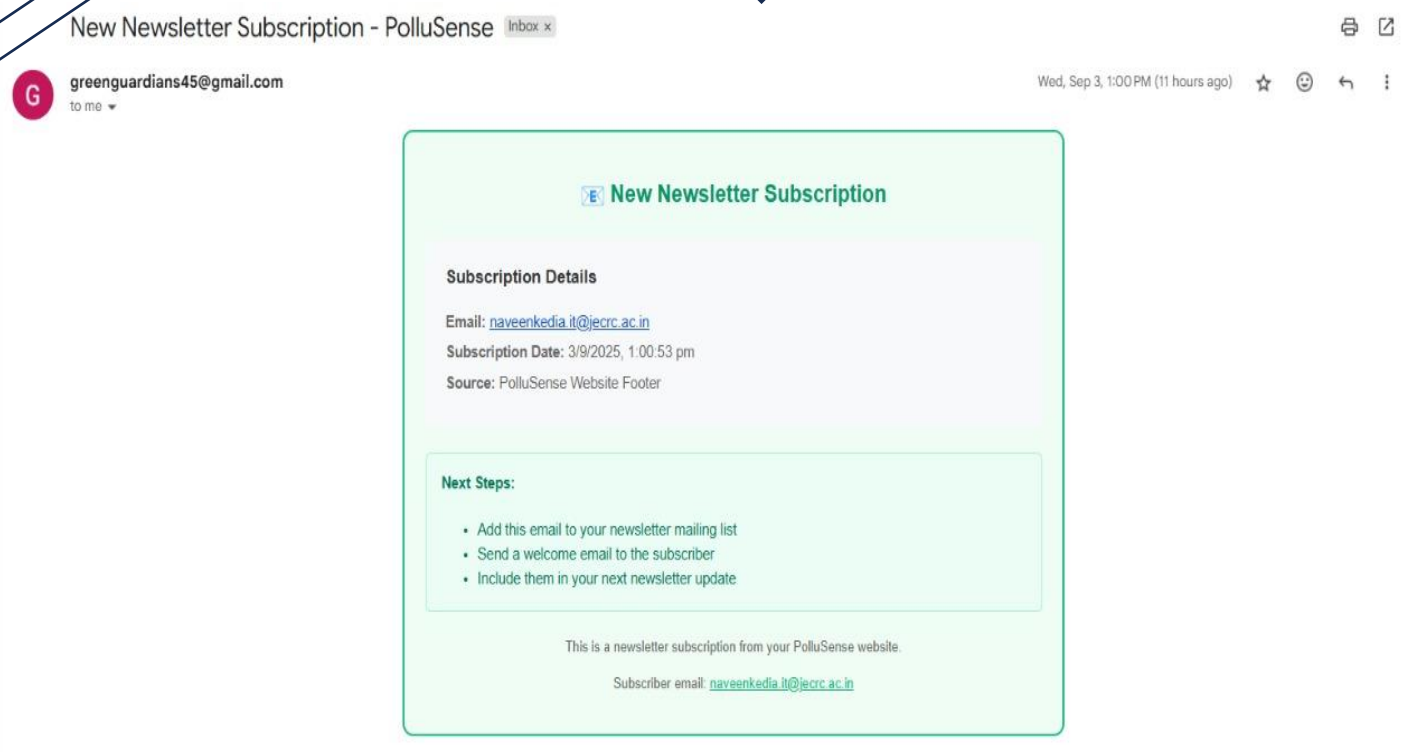

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Enter your full name
EMAIL ADDRESS *
Enter your email address
PHONE NUMBER *
Enter your phone number
COMPANY/ORGANIZATION
Enter your company name
Contact Information
Phone
+91 7307973865
+91 8209949921
Email
GreenGuardians45@gmail.com
support@airquality.ai
Address
JECRC FOUNDATION
Sitapura , Jaipur , 302004



Automated Alert Sent On Mail

NewsLetter Subscription





RESEARCH & REFERENCES

Sl. No.	Feature	Description	Links
1.	AI-Powered Air Quality Monitoring	The integration of AI in monitoring air quality within smart city frameworks.	Artificial intelligence-assisted air quality monitoring for smart city management
2.	IoT-Based Air Pollution Monitoring	This resource offers insights into building mobile-based location systems using GPS and network technologies.	An IoT based low cost air pollution monitoring system
3.	Crowdsourced Pollution Tracking & Public Reporting	This article discusses how crowdsourcing pollution data can enhance public health by involving communities in environmental monitoring. By collecting and sharing pollution information, citizens can contribute to a larger dataset that aids in understanding and mitigating pollution's impact on health.	Phys.org: Crowdsourcing Pollution Data
4.	AI Predictions for Pollution Trends	Researchers at Johns Hopkins University Applied Physics Laboratory are leveraging artificial intelligence to simulate atmospheric models, enhancing the resolution and speed of air quality forecasts. This approach aims to provide earlier and more accurate predictions of air pollution threats, such as wildfire plumes.	Johns Hopkins APL: AI Improves Air Quality Forecasting
5.	Environment Protection Act, 1986	This comprehensive act empowers the Indian government to protect and improve environmental quality, control and reduce pollution from all sources, and prohibit or restrict the setting and operation of any industrial facility on environmental grounds.	Environment Protection Act, 1986
6.	WHO & Economic Impact of Air Pollution	WHO reports air pollution causes 7 million premature deaths annually worldwide. India's annual economic loss due to air pollution is estimated at $150 billion.	WHO: Air Pollution & Health Impact World Bank Report on Economic Costs of Pollution
7.	Video Demo	This is Video Link of working Project	Click here to see the video